A SELF 2-MORPHISM OF THE RINGED TOPOS OF AN ALGEBRAIC SPACE IS THE IDENTITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 04IC: sources/stacks-chow/tags.txt, source lines 5898–5898. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03I5: sources/stacks-chow/tags.txt, source lines 4591–4591. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02WL: sources/stacks-chow/tags.txt, source lines 3808–3808. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04JS: sources/stacks-chow/tags.txt, source lines 5948–5948. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00WS: sources/stacks-chow/tags.txt, source lines 1194–1194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X, Y be algebraic spaces over S. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let t be a 2-morphism from $(f_{small}, f^\sharp)$ to itself, see Modules on Sites, Definition 04IC. Then $t = id$.*

Proof. Let X', resp. Y' be X viewed as an algebraic space over $\mathrm{Spec}(\mathbf{Z})$, see Spaces, Definition 03I5. It is clear from the construction that $(X_{small}, \mathcal{O})$ is equal to $(X'_{small}, \mathcal{O})$ and similarly for Y. Hence we may work with X' and Y'. In other words we may assume that $S = \mathrm{Spec}(\mathbf{Z})$.

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Assume $S = \mathrm{Spec}(\mathbf{Z})$, $f : X \to Y$ and t are as in the lemma. This means that $t : f_{small}^{-1} \to f_{small}^{-1}$ is a transformation of functors such that the diagram

$$\begin{array}{ccc} f_{small}^{-1}\mathcal{O}_Y & \xleftarrow{\quad t \quad} & f_{small}^{-1}\mathcal{O}_Y \\ & {\scriptstyle f^\sharp} \searrow \quad \swarrow {\scriptstyle f^\sharp} & \\ & \mathcal{O}_X & \end{array}$$

is commutative. Suppose $V \to Y$ is étale with V affine. Write $V = \mathrm{Spec}(B)$. Choose generators $b_j \in B$, $j \in J$ for B as a $\mathbf{Z}$-algebra. Set $T = \mathrm{Spec}(\mathbf{Z}[\{x_j\}_{j\in J}])$. In the following we will use that $\mathrm{Mor}_{Sch}(U, T) = \prod_{j\in J} \Gamma(U, \mathcal{O}_U)$ for any scheme U without further mention. The surjective ring map $\mathbf{Z}[x_j] \to B$, $x_j \mapsto b_j$ corresponds to a closed immersion $V \to T$. We obtain a monomorphism

$$i : V \longrightarrow T_Y = T \times Y$$

of algebraic spaces over Y. In terms of sheaves on $Y_{étale}$ the morphism i induces an injection $h_i : h_V \to \prod_{j\in J} \mathcal{O}_Y$ of sheaves. The base change $i' : X \times_Y V \to T_X$ of i to X is a monomorphism too (Spaces, Lemma 02WL). Hence $i' : X \times_Y V \to T_X$ is a monomorphism, which in turn means that $h_{i'} : h_{X\times_Y V} \to \prod_{j\in J} \mathcal{O}_X$ is an injection of sheaves. Via the identification $f_{small}^{-1} h_V = h_{X\times_Y V}$ of Lemma 04K2 the map $h_{i'}$ is equal to

$$f_{small}^{-1} h_V \xrightarrow{f^{-1}h_i} \prod_{j\in J} f_{small}^{-1}\mathcal{O}_Y \xrightarrow{\prod f^\sharp} \prod_{j\in J} \mathcal{O}_X$$

(verification omitted). This means that the map $t : f_{small}^{-1} h_V \to f_{small}^{-1} h_V$ fits into the commutative diagram

$$\begin{array}{ccccc} f_{small}^{-1} h_V & \xrightarrow{f^{-1}h_i} & \prod_{j\in J} f_{small}^{-1}\mathcal{O}_Y & \xrightarrow{\prod f^\sharp} & \prod_{j\in J} \mathcal{O}_X \\ \downarrow{\scriptstyle t} & & \downarrow{\scriptstyle \prod t} & & \downarrow{\scriptstyle \mathrm{id}} \\ f_{small}^{-1} h_V & \xrightarrow{f^{-1}h_i} & \prod_{j\in J} f_{small}^{-1}\mathcal{O}_Y & \xrightarrow{\prod f^\sharp} & \prod_{j\in J} \mathcal{O}_X \end{array}$$

The commutativity of the right square holds by our assumption on t explained above. Since the composition of the horizontal arrows is injective by the discussion above we conclude that the left vertical arrow is the identity map as well. Any sheaf of sets on $Y_{étale}$ admits a surjection from a (huge) coproduct of sheaves of the form h_V with V affine (combine Lemma 04JS with Sites, Lemma 00WS). Thus we conclude that $t : f_{small}^{-1} \to f_{small}^{-1}$ is the identity transformation as desired. □